

SETTLO

Data Model — Prisma Schema
PostgreSQL 18 | Prisma 7 | POC Version 1.0 | May 2026

Design Decisions

- **Multi-entity:** One User can own multiple BusinessEntity records — architecture-ready from day one.
- **Accountants as users:** Role ACCOUNTANT is a first-class User with their own login and assignments.
- **Canton config in DB:** All fiscal rates (AHV, IV, EO, federal, cantonal, communal, VAT) stored in DB with effectiveFrom/To versioning. Never hardcoded.
- **Decimal(18,8) everywhere:** All monetary amounts use Decimal(18,8) — never Float. Supports fiat and crypto.
- **Currency-ready:** CHF only for POC. Currency + ExchangeRate models ready for EUR, USD, BTC, ETH without schema changes.
- **Soft deletes:** Critical entities (User, BusinessEntity, Client, Invoice, Expense) use deletedAt instead of hard delete.
- **AI flywheel:** Every accountant answer captured in KnowledgeBaseEntry with pgvector-ready embedding field for RAG.
- **Audit trail:** TaxEstimation stores a ratesSnapshot JSON — calculations remain auditable even after rate changes.

Entity Overview

User — Core auth entity. Roles: OWNER | ACCOUNTANT | ADMIN. Soft delete.

UserSession — Refresh token management for JWT sessions.

Subscription — Stripe integration, trial tracking, monthly human answer quota.

BusinessEntity — The self-employed business. One User can own many. Canton-linked.

AccountantAssignment — Links an ACCOUNTANT user to a BusinessEntity.

Canton — 26 Swiss cantons with all fiscal rates. Versioned with effectiveFrom/To.

Commune — Communal tax multipliers per canton. BFS number as unique key.

Currency — CHF active for POC. FIAT + CRYPTO types. Decimals field for precision.

ExchangeRate — Historical rates for CHF pairs. Source tracked (SNB, CoinGecko).

Client — Invoice recipients. Full QR-bill address fields per SIX standard.

Invoice — QR-bill compliant. Status machine. Reminder tracking. Soft delete.

InvoiceLineItem — Line items with quantity, unit price, VAT rate.

Payment — Manual payment recording against invoices.

Expense — Receipt upload, OCR data, AI confidence, user override tracking.

ExpenseCategory — Swiss tax categories with default deductibility rules.

TaxEstimation — Core differentiator. AHV+IV+EO+federal+cantonal+communal+VAT.

AiConversation — Container for a chat thread between User and AI.

AiMessage — Individual message with model, confidence, token usage.

AiEscalation — User requests accountant verification. SLA tracked.

KnowledgeBaseEntry — Every accountant answer stored. pgvector-ready for RAG.

BankAccount — IBAN storage. bLink sync in v2.

Notification — Push/in-app notifications with typed payload.

Full Schema (prisma)

```
// =====
// SETTLO – Prisma Schema
// PostgreSQL 18 | Prisma 7
// Version: 1.0 POC
// =====
// Design decisions:
// - Multi-entity from day one: User → (many) BusinessEntity
// - Accountants are first-class users with role ACCOUNTANT
// - Canton fiscal config stored in DB (never hardcoded)
// - Currency-ready: all monetary amounts stored as Decimal(18,8)
//   to support fiat and crypto. CHF only for POC.
// - Soft deletes on critical entities (deletedAt)
// - All timestamps in UTC
// =====

generator client {
  provider      = "prisma-client-js"
  previewFeatures = ["postgresqlExtensions"]
}

datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
  extensions = [uuidOssp(map: "uuid-oss"), pgcrypto]
}

// =====
// ENUMS
// =====

enum UserRole {
  OWNER      // Self-employed person, owns one or more BusinessEntity
  ACCOUNTANT // Certified accountant, can be assigned to BusinessEntity
  ADMIN      // Settlo internal admin
}

enum UserStatus {
  ACTIVE
  SUSPENDED
  PENDING_VERIFICATION
}

enum SubscriptionTier {
  SOLO      // CHF 19/mo – software only, no human accountant access
  PRO       // CHF 49/mo – tax engine + 1 human answer/mo
  CONFIDENCE // CHF 99/mo – full service, 3 human answers/mo + annual review
}

enum SubscriptionStatus {
  TRIALING
  ACTIVE
  PAST_DUE
  CANCELED
  UNPAID
}

enum BusinessEntityType {
  SOLE_PROPRIETORSHIP // Einzelunternehmen – primary POC target
  GmbH                // Future
  AG                   // Future
}
```



```
// POC: manual/CSV import only – bLink integration in v2
// =====

model BankAccount {
  id          String    @id @default(dbgenerated("uuid_generate_v4())) @db.Uuid
  businessEntityId String @db.Uuid
  bankName    String
  iban        String
  accountName String?
  currencyCode String    @default("CHF")
  isDefault   Boolean    @default(false)
  lastSyncAt  DateTime?

  createdAt    DateTime @default(now())
  updatedAt    DateTime @updatedAt

  businessEntity BusinessEntity @relation(fields: [businessEntityId], references: [id], onDelete: Cascade)

  @@index([businessEntityId])
  @@map("bank_accounts")
}

// =====
// NOTIFICATIONS
// =====

model Notification {
  id          String    @id @default(dbgenerated("uuid_generate_v4())) @db.Uuid
  userId      String    @db.Uuid
  type        String    // "invoice_overdue" | "vat_threshold" | "escalation_answered" | etc.
  title       String
  body        String
  data        Json?     // Extra context (invoiceId, escalationId, etc.)
  readAt      DateTime?
  createdAt   DateTime  @default(now())

  user User @relation(fields: [userId], references: [id], onDelete: Cascade)

  @@index([userId, readAt])
  @@map("notifications")
}
```

Settlo AG — Confidential. Copy this file to your NestJS project as `prisma/schema.prisma` and run: `npx prisma migrate dev --name init`